

Bogoan Kim, Ph.D.

CONTACT INFORMATION	Building E8-10, Room 302 Chungdae-ro 1, Seowon-gu, Cheongju-si, Chungcheongbuk-do, 28644, Korea	Tel: + 82 - 43 - 261 - 2483 Email: bogoan@cbnu.ac.kr Homepage: https://bogoankim.github.io/
RESEARCH INTEREST	My research sits at the intersection of human-computer interaction (HCI) , digital health , and inclusive AI . I design and empirically evaluate AI-augmented systems—spanning mobile applications, immersive (VR) environments, and LLM-powered tools—grounded in multimodal behavioral, physiological, and contextual data. A particular emphasis is placed on preserving human agency in AI-driven processes and on the meaningful integration of AI into real-world workflows across diverse professional and social contexts, informed by participatory design and evidence synthesis.	
POSITIONS	Chungbuk National University , Cheongju, Korea • <i>Assistant Professor, School of Information and Communication Engineering</i> • <i>Affiliated Researcher, Cancer Research Center</i> • <i>Director, CBNU HCI Lab</i> KAIST , Daejeon, Korea • <i>Jang Young-Sil Postdoctoral Fellow, College of Engineering</i> – Host Professors: Profs. Hwajung Hong and Uichin Lee The Pennsylvania State University , PA, USA • <i>Visiting Scholar, College of IST</i> – Mentors: Profs. Dongwon Lee and Aiping Xiong Indiana University , IN, USA • <i>Visiting Scholar, Luddy School of Informatics</i> – Mentors: Profs. Patrick C. Shih and Georgia C. Frey	Sep 2025 – Present Mar 2025 – Aug 2025 Jan 2020 – Jun 2020 Jun 2019 – Sep 2019
EDUCATION	Hanyang University , Seoul, Korea • <i>Ph.D. in Artificial Intelligence</i> – Dissertation: BalancePath: AI-VR Framework for Personalized Adaptive System – Advisor: Prof. Kyungsik Han University of Minnesota – Twin Cities , MN, USA • <i>B.S. in Statistics (Graduated with honors)</i> – Thesis: Independent Associations of MVPA with Blood Pressure in Hypertensive Adults – Thesis Supervisor: Prof. Barbara Kuzmak	
AWARDS & HONORS	ACM CHI 2026 Best Paper Honorable Mention • Awarded for two publications, [c.10] and [c.12] (Top 5% of submissions) Jang Young Sil Fellowship • National fellowship for outstanding researchers under the MSIT • Proposal title: Personalized Discomfort Prediction via Multimodal LLMs NRF of Korea Ph.D. Fellowship • Granted 40M KRW (equivalent to 35,000 USD) stipend for 2-year Ph.D. studies • Selected among highly competitive STEM applicants nationwide (Top 180 in Open Track) • Proposal title: AI+VR-based QoLT Systems for Social Independence of Autistic Individuals SIGCHI Travel Grant for UIST 2023 Doctoral Symposium Gary Marsden Travel Award for CHI 2023 SK Creative Challenge Finalist IITP Best Global Research Team Award • Selected as one of 15 teams out of 382 (Top 3.9%) • Proposal title: Human Factors in Fake News Detection and Echo Chamber Mitigation	2026 2025 2023 2023 2023 2022 2021

IITP High-Potential Individuals Global Training Fellowship	2020
• Support for research @ Pennsylvania State University	
Indiana University Global Talent Attraction Program (GTAP) Fellowship	2019
• Support for research @ Indiana University	
Honors on the Dean's List	2011, 2012, 2015
Global Excellence Scholarship (merit-based; 4,210 USD)	2011
Special Recognitions for Outstanding Reviews	
• CHI 2024, CHI 2025, IMWUT 2025, CHI 2026	
Domestic Conference Awards	
• Grand Paper Award: KDMS 2024	
• Best Paper Award: KDMS 2024	
• Best Presentation Award: KCC 2024	

PUBLICATIONS

Conference Proceedings (★: [NRF top-tier CS conferences](#); *: equal contribution)

[c.13] When Special Education Meets LLMs: Investigating the use of LLM-based Conversational AI by Special Education Teachers for Autism in China

J Wu, D Choi, [B Kim](#), and H Hong

DIS 2026 (*ACM International Conference on Designing Interactive Systems*)

Best Paper Honorable Mention Award (Top 5%)

[c.12] QuerySwitch: Supporting the Design Process by Balancing Vagueness through Large Language Models

★ M Kim, [B Kim](#), and K Han

CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#) [10-min Talk](#) [Slides](#) [Media](#)

[c.11] LAPS: Automating Hypothesis-Driven Statistical Analysis of Public Survey Using Large Language Models

★ {J Kim*, D Jeong*}, B Son, H Kim, [B Kim](#), and K Han

CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#) [10-min Talk](#) [Slides](#) [Media](#)

Best Paper Honorable Mention Award (Top 5%)

[c.10] “I Choose to Live, for Life Itself”: Understanding Agency of Home-Based Care Patients Through

★ Information Practices and Relational Dynamics in Care Networks

{S-I Kim*, J Park*}, [B Kim](#), and H Hong

CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#)

[c.9] Understanding Human-Multi-Agent Team Formation for Creative Work

★ H Lim, D Choi, S Nam, [B Kim](#), and H Hong

CHI 2026 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#)

[c.8] “I Don’t Know Why I Should Use This App”: Holistic Analysis on Engagement Challenges in Mental Health

★ {S Jin*, [B Kim*](#)}, and K Han

CHI 2025 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#) [Media](#)

[c.7] Narrating Routines through Game Dynamics: Impact of a Routine Management App for Autistic Individuals

★ [B Kim](#), D Jeong, H Hong, and K Han

CHI 2024 (*ACM CHI Conference on Human Factors in Computing Systems*)

Links : [PDF](#) [10-min Talk](#) [System Teaser](#)

[c.6] V-DAT: Supporting Self-Awareness for Autistic People from Multimodal VR Sensor Data

★ [B Kim](#), D Jeong, JG Kim, H Hong, and K Han

UIST 2023 (*ACM Symposium on User Interface Software and Technology*)

Links : [PDF](#) [10-min Talk](#) [System Teaser](#)

- [c.5] Supporting Independence of Autistic Adults through Mobile and Virtual Reality Technologies
 * B Kim (Doctoral Symposium)
UIST 2023 (*ACM Symposium on User Interface Software and Technology*)
 Doctoral Symposium
 Links : [PDF](#)
- [c.4] RoutineAid: Externalizing Key Design Elements to Support Daily Routines of Individuals with Autism
 * B Kim, S-I Kim, S Park, HJ Yoo, H Hong, and K Han
CHI 2023 (*ACM CHI Conference on Human Factors in Computing Systems*)
 Links : [PDF](#) [10-min Talk](#) [System Teaser](#)
- [c.3] VISTA: User-Centered VR Training System for Effectively Deriving Characteristics of Autistic Individuals
 * B Kim, D Jeong, M Jeong, T Noh, S-I Kim, T Kim, S-Y Jang, HJ Yoo, JG Kim, H Hong, and K Han
VRST 2022 (*The ACM Symposium on Virtual Reality Software and Technology*)
 Links : [PDF](#) [10-min Talk](#) [System Teaser](#) [Media](#)
- [c.2] ChamberBreaker: Mitigating the Echo Chamber Effect through a Gamified Inoculation System
 * Y Jeon, B Kim, A Xiong, D Lee, and K Han
CSCW 2021 (*ACM Conference on Computer-Supported Cooperative Work and Social Computing*)
 Links : [PDF](#) [Media](#)
- [c.1] Photos Don't Have Me, But How Do You Know Me?: Analyzing and Predicting Users on Instagram
 K Han, Y Jo, Y Jeon, B Kim, J Song, and S-W Kim
UMAP 2018 (*ACM Conference on User Modeling, Adaptation and Personalization*)
 Links : [PDF](#)

Journal Articles (*: equal contribution)

- [j.11] VR-ACT: Transforming Clinical Evidence of ACT through VR for Subthreshold Depression
B Kim, D Jeong, YS Choi, H Kim, and K Han
Springer Virtual Reality (*SCIE; IF 5.0; JCR 11.2%; Published 2026.03*)
 Links : [PDF](#)
- [j.10] Mobile and VR Technologies for Social Independence in Autistic Individuals: A Scoping Review
B Kim, D Kang
Journal of Korean Institute of Intelligent Systems (*KCI; Published 2026.03*)
- [j.9] Enhancing Mindfulness-Based Cognitive Therapy in a Virtual Reality: A Prospective Interventional Study
 {B Kim*, D Jeong*}, YS Choi, Y Choi, H Kim, and K Han
Scientific Reports (*SCIE; IF 3.9; JCR 18.0%; Published 2025.09*)
 Links : [PDF](#)
- [j.8] Development of a Personalized Early Prediction Model for Cybersickness in Virtual Reality
 YS Choi, D Jeong, B Kim, and K Han
KIISE Transactions on Computing Practices (*KCI; Published 2025.06*)
- [j.7] Promoting Self-Efficacy of Individuals with Autism in Practicing Social Skills in the Workplace Using Virtual Reality and Physiological Sensors: Mixed Methods Study
 S-I Kim, S-Y Jang, T Kim, B Kim, D Jeong, T Noh, M Jeong, HJ Yoo, K Han, H Hong, and JG Kim
JMIR Formative Research (*SCIE; IF 2.20, Published 2024.01*)
 Links : [PDF](#) [Video](#)
- [j.6] A Study on Mobile Game Design to Support Daily Routines of Individuals with Autism
B Kim, S-I Kim, S Park, HJ Yoo, H Hong, and K Han
Journal of the HCI Society of Korea (*KCI; Published 2022.05*)
- [j.5] Usability Assessment of a Mobile Health Intervention for Increasing Physical Activity in Autistic Adults
 DH Lee, GC Frey, A Min, B Kim, DJ Cothran, S Bellini, K Han, and PC Shih
Health Informatics Journal (*SCIE; IF 2.3; Published 2020.09*)
 Links : [PDF](#)

- [j.4] PuzzleWalk: A Theory-Driven Iterative Design of a Mobile Game for Promoting Physical Activity in Autistic Adults
B Kim, DH Lee, A Min, S Paik, GC Frey, S Bellini, K Han, and PC Shih
PLoS ONE (SCIE; IF 2.9; Published 2020.09)
Links : [PDF](#)
- [j.3] No More One Liners: Bringing Context into Emoji Recommendations
J-G Kim, T Gong, B Kim, J Park, W Kim, E Huang, K Han, J Kim, JG Ko, and S-J Lee
ACM Transactions on Social Computing 2020
Links : [PDF](#)
- [j.2] A Systematic Review on Fake News Research through the Lens of News Creation and Consumption: Research Efforts, Challenges, and Future Directions
B Kim, A Xiong, D Lee, and K Han
PLoS ONE (SCIE, IF = 2.9, Published 2020.09)
Links : [PDF](#)
- [j.1] ATM: The Effectiveness of Self-Regulation on Time Management through a Smartphone Application
B Kim, S-W Lee, H Hong, and K Han
IEEE Access (SCIE, IF = 3.4, Published 2019.07)
Links : [PDF](#)

Posters and Workshops

- [p.9] Agreement Between Smartphone- and Research-grade Accelerometer-measured Step Counts in Autistic Adults in Free-living Conditions
L McNulty, **B Kim**, AE Bodde, ML Mauriello, BC Helsel, and D Lee
ACSM 2026 (*American College of Sports Medicine*)
- [p.8] Early Prediction of Cybersickness in Virtual Reality Using an LLM for Multimodal Time Series Data
YS Choi, D Jeong, **B Kim**, and K Han
UbiComp 2024 (*ACM Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies*)
- [p.7] Objectively Measured Physical Activity and Sedentary Time in Autistic Adults
D Lee, PC Shih, **B Kim**, K Han, A Min, D Cothran, S Bellini, and GC Frey
ACSM 2020 (*American College of Sports Medicine*)
- [p.6] A Study on the Text-Based Characteristics of Autistic Individuals through Social Media Data Analysis
J Han, **B Kim**, and K Han
ICNGC 2020 (*The International Conference on Next Generation Computing*)
- [p.5] FashionQ: An Interactive Tool for Analyzing Fashion Style Trend with Quantitative Criteria
Y Jeon, S Jin, **B Kim**, and K Han
CHI 2020 (*ACM Conference on Human Factors in Computing Systems*)
- [p.4] Ecological Momentary Assessment of Physical Activity and Anxiety in Autistic Adults
D Lee, D Cothran, S Bellini, PC Shih, K Han, **B Kim**, A Min, S Paik, and GC Frey
INSAR 2020 (*International Society for Autism Research*)
- [p.3] Bringing Context into Emoji Recommendations
J-G Kim, T Gong, **B Kim**, J Park, E Huang, W Kim, K Han, J Kim, JG Ko, and S-J Lee
MobiSys 2019 (*ACM Conference on Mobile Systems, Applications, and Services*)
- [p.2] Iterative Design and Usability of Gamified Application for Adults with Autism Spectrum Disorder
S Paik, **B Kim**, and K Han
ICNGC 2019 (*The International Conference on Next Generation Computing*)
- [p.1] Leveraging Mobile Technology to Support Goal Setting and Strategies of College Students
B Kim, S Rhim, K Han, and S-W Lee
UbiComp 2018 (*ACM Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies*)

PATENTS
& SOFTWARE

Granted Patents

- VR-based Interactive Social Skills Training System for Autistic Individuals
Registration Number: **KR10-2938949** Mar 2026
- A Mobile System for Routine Management in Autistic Individuals
Registration Number: **KR10-2729263** Oct 2024

Filed Patents

- A Multimodal LLM Capturing Inter-modal Dynamics for Personalized VR Sickness Prediction
Application Number: **KR10-2025-0100532** Jun 2025
- VR Data Analysis Tool to Support Self-Awareness for Autistic Individuals
Application Number: **KR10-2023-0116955** Aug 2023

Registered Software

- VR-based Dashboard for Clinical Support and Patient Monitoring in Depression Care
Registration Number: **C-2024-034698** Oct 2024

TEACHING
EXPERIENCE

Faculty Instructor (*: courses taught in English)

- [GR] Special Topics in Deep Learning* (8836552-01) Spring 2026
- [UG] Information Technology Programming (0914006-10) Spring 2026
- [UG] Gaesin Frontier II (0281087-12) Spring 2026
- [UG] Capstone Design (5108102-03) Spring 2026
- [UG] Advanced Computer Programming (5108113-01) Spring 2026
- [UG] Intelligent System (5108115-01) Fall 2025
- [UG] Object-Oriented Programming (5108124-01) Fall 2025
- [UG] Fundamental Computer Programming (0914002-18) Fall 2025

Instructor / Co-lecturer (*: courses taught in English)

- Computational Social Science* Spring 2025
- Human-Computer Interaction* Fall 2024
- Human-Computer Interaction* Fall 2023
- Introduction to Software Design* Spring 2023
- Creative Software Design* Fall 2022
- Introduction to Software Design* Spring 2022
- Creative Software Design* Fall 2021
- Data Structures and Practice* Spring 2021
- AI Intensive Class Fall 2019
- Data Structures and Practice* Spring 2019

Teaching Assistant

- Algorithms Fall 2020
- Introduction to AI Programming Language Spring 2020
- The Basics of Data Analytics Fall 2018
- The Basics of Data Analytics Spring 2018

INVITED TALKS

Conference Invited Talk, Dajeon, Korea

- Host: Korean Society of Veterinary Science Spring Conference 2026
- Session: The Next-Generation Research Paradigm in the Era of AI Transformation (AX)
- Topic: Designing Agency-Centered Interactions in Digital Health Apr 2026

Lab Guest Speaker Session, Seoul, Korea

- Host: MINDS Lab @ Chung-Ang University
- Topic: Designing Sociotechnical Systems for People in Need Aug 2025

Europe-Korea Conference 2025 Seminar, Seoul, Korea

- Host: College of Electrical & Computer Engineering @ Chungbuk National University
- Topic: The Role of HCI Researchers in the Development of Digital Therapeutics Aug 2025

Lab Guest Speaker Session , Daejeon, Korea	May 2025
• Host: DxD Lab @ KAIST	
• Topic: AI-Infused Software Systems for Solving Real-World Challenges	
Omnibus Lecture Series on Computational Social Science , Seoul, Korea	Mar 2025
• Host: Department of Data Science @ Hanyang University	
• Topic: User-Centered Digital Healthcare	
Computational Social Science Colloquium , Seoul, Korea	May 2024
• Host: Center for Computational Social Science @ Hanyang University	
• Topic: Sociotechnical Systems for People Who Are in Need	
Top Conference Session Invited Talk , Hongcheon, Korea	Jan 2024
• Host: HCI Korea 2024	
• Topic: Impact of a Gamified Routine Management App for Autistic Adults	
UIST Doctoral Symposium , San Francisco, CA USA	Oct 2023
• Host: ACM UIST 2023	
• Topic: AI-VR Framework for Personalized Adaptive System	
Top Conference Session Invited Talk , Jeju, Korea	Jun 2023
• Host: Korea Computer Congress (KCC) 2023	
• Topic: Externalizing Key Design Elements to Support Daily Routines of Autistic Adults	
IITP Global Talent Program Workshop , Seoul, Korea	Dec 2021
• Host: Institute of Information & Communications Technology Planning & Evaluation (IITP)	
• Topic: Explainable Fake News Detection with Deep Learning and Human Factors	
HCI Korea Doctoral Colloquium , Seoul, Korea	Jan 2021
• Host: HCI Korea 2021	
• Topic: Designing Inclusive Mobile Applications for Marginalized Populations	
Lab Guest Speaker Session , PA, USA	Feb 2020
• Host: The PIKE Group @ The Pennsylvania State University	
• Topic: From Creation to Consumption: Data-Driven Perspectives on Fake News Research	
Invited Guest Lecture , Suwon, Korea	Oct 2019
• Host: AI Intensive Class @ Ajou University	
• Topic: Designing a Gamified mHealth App for Marginalized Populations	
Global Talent Attraction Program Conference , IN, USA	Jun 2019
• Host: Luddy School of Informatics, Computing, and Engineering @ Indiana University	
• Topic: Designing a Mobile Game to Promote Physical Activity in Autistic Adults	
Global Challenge Program Symposium , Tokyo, Japan	Jun 2018
• Host: Environment and Information Studies @ Keio University	
• Topic: Effectiveness of a Smartphone App in Self-Regulating Time Management	

ACADEMIC SERVICES

Conference Reviewer	
• The Conference on Human Factors in Computing Systems (CHI)	2019 - 2026
• The Conference on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)	2019 - 2025
• The Designing Interactive Systems Conference (DIS)	2025
• The Conference on Computer-Supported Cooperative Work & Social Computing (CSCW)	2019 - 2024
• The Conference on Information and Knowledge Management (CIKM)	2021 - 2023
• The Symposium on Applied Computing (SAC)	2022
• The Symposium on Mixed and Augmented Reality (ISMAR)	2021 - 2022
Journal Reviewer	
• npj Digital Medicine	2026
• ACM Transactions on Accessible Computing	2026
• JMIR Formative Research	2025
• The Nature Scientific Reports	2024
• The Taylor and Francis Interactive Learning Environments	2023

Student Volunteer

- The Conference on Mobile Systems, Applications, and Services (**MobiSys**)

2019

ACADEMIC MENTORSHIP

Mentored Students

- C Lee, Chungbuk National University (Master's Student) Fall 2025 - Present
- S Seo, Chungbuk National University (Undergrad. Student) Fall 2025 - Present
- D Jeong, Chungbuk National University (Undergrad. Student) Fall 2025 - Present
- D Lee, Chungbuk National University (Undergrad. Student) Fall 2025 - Present
- J Kim, Hanyang University (Ph.D. Student) Spring 2025 - Spring 2025
- L McNulty, University of Delaware (Ph.D. Student) Spring 2025 - Fall 2025
- H Lim, KAIST (Ph.D. Student) Spring 2025 - Fall 2025
- S-I Kim, Seoul National University (MD, Ph.D. Student) Fall 2024 - Fall 2025
- J Park, KAIST (Master's Student) Spring 2025 - Fall 2025
- J Wu, KAIST (Ph.D. Student) Spring 2025 - Fall 2025
- M Kim, Hanyang University (Ph.D. Student) Spring 2024 - Present
- YS Choi, Hanyang University (Master's Student) Fall 2023 - Present
- D Jeong, Hanyang University (Ph.D. Student) Spring 2022 - Present
- S Paik, Ajou University (Master's Student) Spring 2019 - Fall 2020

PRESS & INTERVIEWS

Research Media Coverage

- "충북대 김보관 교수 연구팀, 우수논문상 2편 수상" Apr 2026
- "인간의 가치를 지키는 'AI 가치 정렬' 연구 성과..." Mar 2026
- "정신건강관리 앱 지속사용 한계..." May 2025
- KAIST 등, 자폐스펙트럼장애 직장 적응 돕는 VR 개발 Jan 2023
- New game can help users identify, avoid online echo chambers Oct 2021

REFERENCES

Hwajung Hong, *Professor* (Postdoc. Host Professor)
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